MILITARY SPECIFICATION

CUTTING FLUIDS, SULFUR-AND CHLORINE-ADDITIVE

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers blended cutting fluids containing sulfur and chlorine additives, with or without fatty oils, in a mineral oil base (see 6.1).

1.2 Classification. The cutting fluids shall be of the following grades, classified according to fatty oil content.

Grade	Fatty oil content: percent, minimum
1	No requirement
2	1.0
3	6.0
4	12.0
5	30.0

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY
MIL-F-46148(MR) - Fatty Oil (for Metal-Working Lubricants)

STANDARDS

FEDERAL
Fed. Test Method Std. No. 791 - Lubricants, Liquid Fuels, and Related Products; Methods of Testing

FSC 9150

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MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-290 - Packaging, Packing, and Marking of Petroleum and Related Products

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) TEST METHODS

D 92 - Flash and Fire Points by Cleveland Open Cup
D 95 - Water in Petroleum and Other Bituminous Materials
D 97 - Pour Point
D 270 - Sampling Petroleum and Petroleum Products
D 445 Viscosity of Transparent and Opaque Liquids (Kinematic and Dynamic Viscosities)
D 808 - Chlorine in New and Used Petroleum Products (Bomb Method)

(The ASTM test methods listed above are included in Parts 17 and 18 of the ASTM Book of Standards and are available individually for sale.)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

Specifications and standards of technical societies are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Material. The cutting fluids shall be blends of fatty oil and refined mineral oil in whatever proportions and with whatever additives that are necessary to meet the requirements of table I.

3.2 Physical and chemical requirements.

3.2.1 Fatty oil blending stock. The fatty oil blending stock shall meet the requirements of MIL-F-46148(MR).

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3.2.2 Mineral oil blending stock. Physical and chemical properties of the mineral oil blended stock shall be such as to permit the blended cutting fluids to meet the requirements of table I.

3.2.3 Blended cutting fluids. Blended cutting fluids shall meet the requirements of table I.

Table I. Physical and chemical requirements for blended cutting fluids

Properties	Values				
	Grade 1	Grade 2	Grade 3	Grade 4	Grade 5
Fatty oils: percent, min.	No req.	1.0	6.0	12.0	30.0
Active sulfur percent, min.	0.5	0.2	1.8	2.0	1.0
Total chlorine, percent, min.	5.5	0.1	0.8	2.7	0.5
Viscosity at 100°F (37.8°C); centistokes	5-9	4-6	35-40	18-22	40-44
Pour point: °F max (°C max)	30 (-1)	30 (-1)	30 (-1)	30 (-1)	30 (-1)
Flash point: °F min (°C min)	255 (124)	285 (140)	285 (140)	285 (140)	285 (140)
Sediment and phase separation	None	None	None	None	None
Water content, percent by volume, max	0.2	0.2	0.2	0.2	0.2

3.3 Toxicity. Cutting fluids shall have no adverse effects on the health of personnel when they are used as intended. Questions pertaining to toxicity shall be referred by the procuring activity to the appropriate departmental medical service (see 6.3) which will act as advisor to the procuring activity (see 4.6.2).

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3.4 Workmanship. The cutting fluids shall be homogeneous, free-flowing, and free from objectionable odors.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Lot.

4.2.1 Bulk lot. An indefinite quantity of a homogeneous blend of one grade of cutting fluid offered for acceptance in a single, isolated container; or manufactured in a single plant run (not exceeding 24 hours) through the same processing equipment with no change in the ingredient materials.

4.2.2 Packaged lot. An indefinite number of drums or other unit containers of identical size and type, offered for acceptance at one time, and filled with a homogeneous blend of one grade of cutting fluid from a single, isolated container; or filled with a homogeneous mixture of one type of concentrate manufactured in a single plant run (not exceeding 24 hours), through the same processing equipment, with no change in the ingredient materials.

4.3 Sampling.

4.3.1 Sampling for examination of filled containers. Take a random sample of filled containers from each lot in accordance with MIL-STD-105, at inspection level II and acceptable quality level (AQL) = 2.5 percent defective.

4.3.2 Sampling for tests. Take samples for tests in accordance with ASTM method D 270.

4.4 Inspection. Perform inspection in accordance with method 9601 of Fed. Test Method Std. No. 791.

4.4.1 Examination of filled containers. Examine samples taken as specified in 4.3.1 for conformance with MIL-STD-290 with regard to fill, closure, sealing, leakage, packaging, packing, and marking requirements. Reject any container having one or more defects or under the required fill. If the number of defective or underfilled containers exceeds the acceptance number for the appropriate sampling plan of MIL-STD-105, reject the lot represented by the sample.

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4.5 Classification of tests. All tests are quality conformance tests.

4.6 Test methods. Perform tests and evaluations in accordance with table II, 4.6.1 and 4.6.2.

Table II. Test methods

Test	Test method No. Fed. Std. 791	Test method No. ASTM
Fatty oil content (see the appendix)		
Active sulfur	3180	
Chlorine		D 808
Viscosity, kinematic		D 445
Pour point		D 97
Flash point		D 92
Sedimentation and phase separation (see 4.6.1)		
Water content		D 95
Toxicity (see 4.6.2)		

4.6.1 Sedimentation and phase separation. Pour a sample of the oil as received into a clean, dry, stoppered 4-oz (30-ml) bottle and allow it to stand for 48 hours at 77° ± 5°F (25° ± 3°C). At the end of the 48-hour period, examine the sample visually for sedimentation or phase separation with the aid of a 40-watt incandescent bulb held approximately six inches behind and below the sample bottle.

4.6.2 Toxicity. The supplier shall furnish the procuring activity with the toxicological data and formulations necessary to evaluate the physiological effects of the product. The data shall include evidence that the product is not injurious to human health when it is used as intended (see 6.1). Such evidence may be in the form of a toxicological study or in some other form acceptable to the preparing activity. Formulations shall include the percentages and standard chemical names of all constituent materials. The procuring activity shall have final authority for determining the acceptability of information or other evidence furnished by a potential supplier relative to the safety of his product.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. Unless otherwise specified in the contract or purchase order (see 6.2), packaging, packing, and marking shall be in accordance with MIL-STD-290.

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6. NOTES

6.1 Intended use. All types are intended for use as received.

6.1.1 Grade 1. For the machining or hand-tapping of all steels and for other materials that are difficult to machine.

6.1.2 Grade 2. For the gun-drilling and trepanning of low carbon or low alloy steel, or the machining or threading of high carbon, high alloy or other steels that are difficult to machine.

6.1.3 Grade 3. For the milling, threading, turning, and hobbing of high carbon steels such as 1025 to 1040 or high alloy steels such as 4130, 4340, and 8620.

6.1.4 Grade 4. For the gun drilling or trepanning of high alloy steels such as 4130, 4340, and 8620.

6.1.5 Grade 5. For the honing of high carbon steels such as 1025 to 1040 or high alloy steels such as 4130, 4340, and 8620.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Grade of cutting fluid required (see 1.2).
c. Quantity of cutting fluid required.
d. Type and size of container (see 5.1).
e. Level of packaging and level of packing required (see 5.1).

6.3 Departmental medical services. Questions concerning the toxicity of commercial cutting fluids offered for sale to the Government should be referred to the appropriate departmental medical service in accordance with 3.3.and 4.6.2. Departmental medical services are as follows:

Army: Office of Surgeon General
Department of the Army
ATTN: MEDDD-SC
Washington, D. C. 20314

Navy: Chief, Bureau of Medicine and Surgery
Bldg. 5, Potomac Annex
Code 4A
Department of the Navy
Washington, D. C. 20390

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Air Force: Office of the Surgeon General
Directorate of Plans and Hospitalization
HQ USAF (AFMSHB)
Forrestal Building
Washington, D. C. 20333

Custodians:
Army - MR
Navy - YD
Air Force - 68

Review activities:
Army - MU, WC
Air Force - 68

User activities:
Army - GL
Navy, SH, OS

Preparing activity:
Army - MR

Project No. 9150-0320

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APPENDIX

DETERMINATION OF FATTY OIL IN CUTTING FLUIDS BY INFRARED SPECTROPHOTOMETRY

10. SCOPE

10.1 This appendix covers a method of analysis for the determination of fatty oil in sulfurized, fatty mineral and sulfurized, chlorinated fatty mineral cutting fluids. The method is based on the assumption that the fatty oil will consist entirely of lard oil. The method is not applicable for the analysis of fatty oils in cutting fluids which contain oxidized oils or compounds containing carbonyl groups which also absorb radiation of 1743 cm^{-1} wave length, the point in the spectrum on which this analysis is based. Compounds containing sulfur and chlorine, which often interfere with the conventional fatty oil analysis do not interfere with this method.

20. APPLICABLE DOCUMENTS

20.1 The following document, of the issue in effect on date of invitation for bids or request for proposal, form a part of this appendix to the extent specified herein.

SPECIFICATIONS

FEDERAL
VV-O-241 - Oil, Cutting, Mineral

30. MATERIALS

30.1 Lard oil meeting the requirements of MIL-F-46148(MR)

30.2 Mineral oil meeting the requirements of grade 2 of VV-O-241.

40. APPARATUS

40.1 An infrared spectrophotometer capable of producing a distinct carbonyl absorption band at 1743 cm^{-1}. The Perkin-Elmer Infrared Spectrophotometer, Model 621, has been found satisfactory. A sealed liquid cell with sodium chloride or potassium bromide windows may be used. It is recommended that the path length used be approximately 0.025 mm.

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50. PROCEDURE

50.1 Calibrate the spectrophotometer in accordance with the manufacturer's instructions. Prepare a series of at least five standards consisting of mixtures of lard oil (see 20.1) and mineral oil (see 20.2) such that the percent of lard oil in the standards ranges from 0.0 to 15.0 percent. If the sample being analyzed contains more than 15 percent lard oil, dilute it with sufficient mineral oil (see 20.1) to reduce the lard oil content below 15 percent so that the standards prepared above can be utilized. Determine the absorbance values of the standards in accordance with 40.2 and plot the percent lard oil against the absorbance values at 1743 cm^{-1}.

50.2 Thoroughly mix the sample to avoid producing air bubbles. Fill the cavity in the sealed cell, avoiding any air inclusions. Place the filled cell in the beam of the infrared spectrophotometer and scan the wave lengths between 2000 and 1500 cm^{-1}. Determine the absorbance of the band at 1743 cm^{-1} using the Base Line Method[1/]. Determine the percentage of lard oil in the sample from the plot prepared from the standards (see 40.1). It is suggested that standards having lard oil concentrations above and below the concentration expected in the test sample be run concurrently with the test sample to check the plot previously prepared from the standards and to compensate for any variability in operation of the apparatus.

60. PRECISION

60.1 Use the following limits as a basis for judging the acceptability of results (95 percent confidence):

a. Repeatability. Duplicate results by the same operator should be considered suspect if they differ by more than 0.2 percent.

b. Reproducibility. The results submitted by each of two laboratories should be considered suspect if they differ by more than 0.3 percent.

1/ Colthup, Daly, and Wiberley, "Introduction to Infrared and Raman Spectroscopy", Academic Press, New York, 1969, page 75.

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